

Original Article

Accurate and efficient structure-based computational mutagenesis for modeling fluorescence levels of *Aequorea victoria* green fluorescent protein mutants

Majid Masso *

Laboratory for Structural Bioinformatics, School of Systems Biology, George Mason University, 10900 University Boulevard MS 5B3, Manassas, VA 20110, USA

*To whom correspondence should be addressed. Email: mmasso@gmu.edu

Edited by: Professor Timothy Whitehead, Board Member, PEDS.

Received 8 March 2020; Revised 16 July 2020; Editorial Decision 4 August 2020; Accepted 4 August 2020

Abstract

A computational mutagenesis technique was used to characterize the structural effects associated with over 46 000 single and multiple amino acid variants of *Aequorea victoria* green fluorescent protein (GFP), whose functional effects (fluorescence levels) were recently measured by experimental researchers. For each GFP mutant, the approach generated a single score reflecting the overall change in sequence-structure compatibility relative to native GFP, as well as a vector of environmental perturbation (EP) scores characterizing the impact at all GFP residue positions. A significant GFP structure–function relationship ($P < 0.0001$) was elucidated by comparing the sequence-structure compatibility scores with the functional data. Next, the computed vectors for GFP mutants were used to train predictive models of fluorescence by implementing random forest (RF) classification and tree regression machine learning algorithms. Classification performance reached 0.93 for sensitivity, 0.91 for precision and 0.90 for balanced accuracy, and regression models led to Pearson's correlation as high as $r = 0.83$ between experimental and predicted GFP mutant fluorescence. An RF model trained on a subset of over 1000 experimental single residue GFP mutants with measured fluorescence was used for predicting the 3300 remaining unstudied single residue mutants, with results complementing known GFP biochemical and biophysical properties. In addition, models trained on the subset of experimental GFP mutants harboring multiple residue replacements successfully predicted fluorescence of the single residue GFP mutants. The models developed for this study were accurate and efficient, and their predictions outperformed those of several related state-of-the-art methods.

Key words: GFP, machine learning, prediction, structure–function relationships

Introduction

Subsequent to the discovery and study of green fluorescent protein (GFP) from the jellyfish *Aequorea victoria* by Osamu Shimomura in the 1960s and 1970s (Morise *et al.*, 1974; Shimomura, 1979; Shimomura *et al.*, 1962), the cloning and determination of the GFP nucleotide sequence were first reported by Prasher *et al.* (1992). As

early as 1987, Prasher envisioned the potential for GFP to be used as a tracer and reporter molecule in other biological systems (Prasher *et al.*, 1986; Prasher *et al.*, 1987), and Martin Chalfie successfully used the clone isolated by Prasher to express GFP in both bacteria and nematodes by 1994 (Chalfie *et al.*, 1994). Key to their success was the fact that GFP could spontaneously form an internal chromophore

in a manner independent of any factor unique to jellyfish, requiring only molecular oxygen (Day and Davidson, 2009). In 1994, Roger Tsien began using experimental mutagenesis techniques to further understand the mechanisms of GFP function (Cubitt *et al.*, 1995; Heim *et al.*, 1994; Heim and Tsien, 1996; Wachter *et al.*, 1997), efforts which were significantly facilitated by 1996 when he and co-workers solved the GFP crystal structure (Ormo *et al.*, 1996). For the next two and a half decades since the discoveries of Prasher and Chalfie, laboratory techniques employing GFP as an *in vivo* fluorescence tag or reporter molecule (Leonetti *et al.*, 2016) in cell and molecular biology, as well as in medicine and physiology, have become ubiquitous and essential; moreover, incremental advancements in live-cell imaging technologies underscore the sustained relevance of GFP in experiments (Day and Davidson, 2009; Tsien, 1998).

A sequence of 238 amino acid residues defines the GFP protein, which folds into an 11-stranded β -barrel wrapped around an interior helix containing S65-Y66-G67 at its center. This tripeptide undergoes spontaneous cyclization and dehydration, contingent upon its orientation within the cylindrical native GFP structure, to yield the mature chromophore (Ormo *et al.*, 1996). A final dehydrogenation step in chromophore development requires molecular oxygen, providing the chromophore with its characteristic absorbance and fluorescence properties (Tsien, 1998). This reaction is catalyzed by residues R96 and E222, which are fully conserved among all fluorescent proteins (Day and Davidson, 2009). Each of the 11 strands forming the cylindrical structure are ~ 9 –13 residues in length and collectively represent the vast majority of the GFP sequence (Ormo *et al.*, 1996). Also, the GFP structure is compact, consisting of tightly packed residue side chains that form a highly stable protein (Day and Davidson, 2009). These properties suggest the improbability of designing a modified GFP of reduced size through large sequence deletions; indeed, fluorescence remains unaffected only for small deletions at the amino- and carboxyl-terminal ends, which are both exposed on the surface of the barrel (Day and Davidson, 2009; Ormo *et al.*, 1996; Tsien, 1998).

For the present study, we used an X-ray crystallographic structure of GFP that was solved at a resolution of 0.9 Å and deposited in the Protein Data Bank (PDB (Berman *et al.*, 2000)), which provides atomic coordinate data for a total of 229 consecutive residue positions (PDB accession code 2wur, chain A (Shinobu *et al.*, 2010)) (Fig. 1a). We implemented a computational mutagenesis technique (Masso and Vaisman, 2008) using this structure to score relative changes to GFP sequence-structure compatibility upon single or multiple residue replacements, as well as to generate a profile quantifying EPs at all residue positions for the mutant GFP protein relative to wild-type. This approach was previously applied to an isolated 35-residue V3 loop fragment from the HIV-1 gp120 envelope protein structure, with a relatively small data set of ~ 1200 V3 sequences each containing multiple mutations relative to the native reference (Masso and Vaisman, 2010a). Here we report a first attempt to apply the technique toward the large-scale modeling of tens of thousands of full-length GFP protein mutants, the vast majority of which each contain multiple amino acid substitutions, and to evaluate how well the effects of epistasis (i.e. non-additive effects of multiple mutations) are implicitly captured. Through the application of fluorescence-activated cell sorting, a recent experimental study measured the function (i.e. fluorescence levels) for these tens of thousands of GFP single and multiple mutants synthesized via random mutagenesis (Sarkisyan *et al.*, 2016). To evaluate the effectiveness with which the computational mutagenesis technique models structural

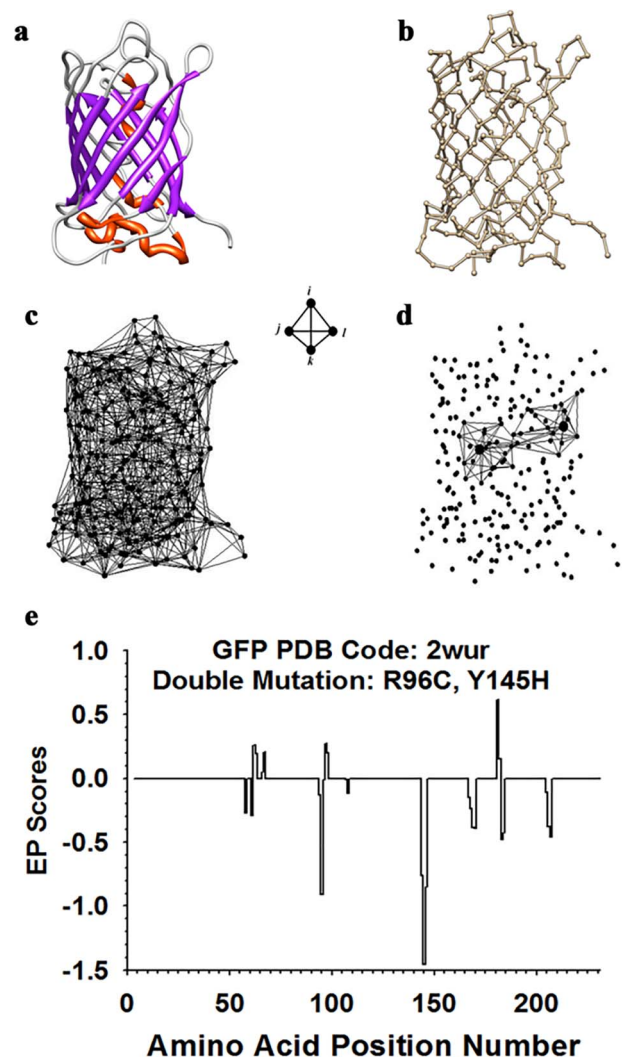


Fig. 1 (a) Ribbon diagram of the *Aequorea victoria* GFP structure from PDB accession code 2wur, chain A. (b) GFP backbone trace obtained using the alpha-carbon atomic coordinates for each of the 229 residues in the solved structure. (c) Delaunay tessellation of the GFP alpha-carbons modified by the removal of edges >12 Å, consisting of 1079 tetrahedra each representing at the four vertices a distinct quadruplet of nearest-neighbor GFP residue positions via their respective alpha-carbons. (d) Amino acid positions R96 and Y145 (two larger points) participate in a total of 44 tetrahedra as shown, which collectively have 25 additional vertices. Residue R94 (resp. Y145) serves as a vertex for 24 (resp. 20) tetrahedra, with 14 (resp. 12) additional residues serving as vertices. Residue T60 is the only common vertex among the two sets of tetrahedral vertices (e) Visual display of the 229-dimensional residual profile used for representing the GFP double mutant R96C, Y145H. The residual score (quantified structural impact) of this mutant is -1.46789 , which is consistent with its measured fluorescence level of 1.33304589132 (functional impact: off).

changes, computed structural data for these GFP mutants were paired with their experimentally derived functional effects. In this regard, sequence-structure compatibility scores for the GFP mutants were used in order to assess the capacity for modeling the GFP structure–function relationship, whereas their environmental profile vectors were used as features to train predictive models of mutant GFP fluorescence with the use of cutting-edge machine learning algorithms.

Materials and Methods

Computational mutagenesis

Underpinning this technique is a four-body knowledge-based statistical potential (Masso and Vaisman, 2010b; Singh *et al.*, 1996), trained using a diverse (<30% sequence identity), high-resolution (<2.2 Å) set of 1417 X-ray crystallographic protein structures (<http://binf.gmu.edu/automute/tessellatable1417.txt>) culled from the PISCES (Wang and Dunbrack, 2003) web server. Proteins were each coarse-grained at the residue level using the 3D coordinates of the alpha-carbons (Fig. 1b), and this point-set was then supplied as input to the Qhull (Barber *et al.*, 1996) implementation of the Delaunay tessellation tiling algorithm from the field of computational geometry (de Berg *et al.*, 2008). Each tiling consisted of solid, non-overlapping, irregular tetrahedra in the shape of a convex hull filling the 3D space occupied by the protein, and all edges >12 Å found on the surface were immediately removed before further analysis (Fig. 1c). The vertices of each tetrahedron in the final tessellation are four alpha-carbons that represent and objectively identify a quadruplet of nearest-neighbor residue positions in the protein structure. A standard 20-letter protein alphabet set can generate 8855 distinct types of such 4-letter amino acid subsets, assuming quadruplets are not ordered and allowing for residue repetitions in any given quadruplet, and relative frequencies of occurrence f were calculated for each quadruplet based on the vertices of several hundred thousand tetrahedra collectively obtained from the 1417 protein structure tessellations. A rate p expected by chance was also calculated for each quadruplet, first by obtaining the observed relative frequencies a_n , $n = 1, 2, \dots, 20$ with which the 20 distinct amino acid types occur in the collective set of 1417 protein structures, then by using the multinomial reference distribution $p = \frac{4!}{\prod_{n=1}^{20} (t_n!)} \prod_{n=1}^{20} a_n^{t_n}$, where t_n is

the number of times the residue of type n appears in the quadruplet. From an application of the inverted Boltzmann principle, an energy of interaction score for each residue quadruplet was computed as $s = \log(f/p)$ (Sippl, 1993; Sippl, 1995), and the collective scores for all 8855 residue quadruplets defined a knowledge-based four-body statistical potential (http://binf.gmu.edu/automute/potential_1417_cut12.txt). Implementation of the four-body potential was performed using an *ad hoc* set of Java and Perl scripts running over the course of 1–2 h.

Using the four-body energy function described above, the ‘total potential’ of any protein with a solved structure can be determined by obtaining its tessellation (subject to removal of edges >12 Å), scoring each tetrahedron as that of the residue quadruplet represented by the alpha-carbons at its four vertices, and adding up these tetrahedral scores (Masso and Vaisman, 2007; Masso and Vaisman, 2008; Masso and Vaisman, 2010b). This computed total potential for a protein is a topological score that quantifies the overall protein sequence-structure compatibility. Since the tessellation forms a packed tiling, such that any adjacent tetrahedral pair may share up to three alpha-carbon vertices, each residue position typically participates in numerous quadruplets of nearest neighbors; moreover, by adding up only the scores of those tetrahedra for which the alpha-carbon of a given amino acid position is shared as a vertex, a residue environment score (RES) is computed for that position (Masso and Vaisman, 2007; Masso and Vaisman, 2008; Masso and Vaisman, 2010b). Collectively, these environment scores for all residue positions in a protein yield a ‘3D-1D potential profile’ vector (Bowie *et al.*, 1991; Masso and Vaisman, 2007). Given any protein mutation defined by a single

or multiple residue replacements, our computational mutagenesis is defined by altering the amino acid labels at the corresponding alpha-carbon vertices in the tessellation of the wild-type protein (Masso and Vaisman, 2010b; Masso and Vaisman, 2008). The scores of all the tetrahedra in the tessellation are recalculated as the total potential and the potential profile for the mutant. To evaluate sequence-structure compatibility changes upon mutation relative to the wild-type as baseline, a mutant ‘residual score’ is defined as the difference between the mutant and wild-type total potentials. Similarly a mutant ‘residual profile’ is defined as the component-wise difference between the potential profiles for the mutant and wild-type proteins. Each component of the mutant residual profile quantifies changes to the local environment at the respective position between the mutant and wild-type protein, and these components are known as EP scores (Fig. 1e).

Mutant GFP data set

For this work, we applied our computational mutagenesis procedure to GFP mutant proteins defined by single and multiple residue replacements distributed among all GFP positions 3–231 from the PDB accession code 2wur, chain A, selected according to those synthesized and reported in the study from which we obtained the experimental mutant fluorescence data (Sarkisyan *et al.*, 2016). In total, fluorescence levels were experimentally measured for 46 310 unique GFP mutants (http://binf.gmu.edu/automute/GFP_mutants_and_measured_fluorescence.txt): 1051 single residue variants and 45 259 multiple mutants each harboring residue replacements occurring at sequence positions ranging from as few as two to as many as 15. Among these GFP mutants, (log) fluorescence values were ≥ 3 (on/light) for 26 309 mutants and < 3 (off/dark) for 20 001 mutants. In particular, there were 953 on/98 off single residue variants and 25 356 on/19 903 off multiple mutants. Residual scores (http://binf.gmu.edu/automute/GFP_mutant_residual_scores.txt) and residual profiles (http://binf.gmu.edu/automute/GFP_mutant_residual_profiles.txt) were computed for all 46 310 mutants.

Machine learning

Each GFP mutant was represented as a 230-dimensional feature vector consisting of its residual profile, whose 229 EP score components form the input attributes (i.e. independent variables or predictors), and its fluorescence as a single output attribute (i.e. dependent variable) in the form of either a numerical value (for regression models) or an on/off categorical variable (for classification models). The computational mutagenesis data (i.e. both the residual scores and residual profiles of the GFP mutants) were obtained by running an *ad hoc* set of Java and Perl scripts over the course of a few hours. Tree regression (REPTree, reduced-error pruned tree) and random forest (RF) classification algorithms were implemented within the Weka software package of machine learning tools under default parameter settings (version 3.6.10) (Frank *et al.*, 2004; Smith and Frank, 2016), in order to train predictive models of GFP fluorescence and to evaluate their performance. For the tree regression implementations in particular, a bagging (i.e. bootstrap aggregating) ensemble procedure available with Weka was applied using default parameters, leading to more accurate predictions being made by averaging those obtained from 10 distinct REPTree models.

Several performance measures were used to evaluate predictions made by the trained RF classification models. Assuming P/N refer to on/off GFP mutants, with TP/TN referring to correctly classified

mutants from each category and FN/FP designating incorrect predictions from the respective classes, we computed

$$se = \text{sensitivity} = TP / (TP + FN),$$
$$sp = \text{specificity} = TN / (TN + FP) \text{ and}$$
$$ppv = \text{positive predictive value (i.e. precision)} = TP / (TP + FP).$$

Moreover, the following reported values combine model performance on both classes:

$$\text{balanced accuracy rate (bar)} = 0.5 \times [se + sp],$$
$$\text{Matthew's correlation coefficient (mcc)} = \frac{(TP \times TN - FP \times FN)}{\sqrt{(TP + FN)(TP + FP)(TN + FN)(TN + FP)}},$$

and the area (auc) under the receiver operating characteristic (ROC) curve, which is a plot of true positive rate (or *se*) versus false positive rate (or $1 - sp$) in the unit square, with auc values that range from 0.5 (i.e. model performs no better than random guessing) to 1.0 (i.e. perfect classifier). Lastly, tree regression models were evaluated by calculating Pearson's correlation coefficient (*r*) between the experimental and predicted fluorescence values for GFP mutants.

Results and Discussion

GFP structure–function relationships

Starting with the individually computed residual scores for the subset of 1051 single residue GFP variants, mean residual scores were calculated for each of the two (on/off) fluorescence categories (Fig. 2a, black ‘All’ bar graph), with the difference between the means being statistically significant (Welch’s *t*-test: $P < 0.0001$) and elucidating a GFP structure (i.e. calculated mean residual scores)—function (i.e. experimentally measured fluorescence) relationship. Within each fluorescence category, the GFP mutants were further segregated based on whether they each represented a conservative (C) or non-conservative (NC) replacement of the native amino acid residue (Fig. 2a, gray and white bar graphs, respectively). With the 20-letter protein alphabet grouped together according to physicochemical similarities as [(A, S, T, G, P), (D, E, N, Q), (R, K, H), (F, Y, W), (V, L, I, M), (C)], replacement of a native residue with another from within the same subgroup was considered to be a conservative mutation, whereas interclass substitutions were considered to be non-conservative (Dayhoff *et al.*, 1978). Note how the non-conservative subsets of GFP mutants in the two fluorescence categories drive the overall structure–function relationship; moreover, conservative residue substitutions minimally impact sequence–structure compatibility on average, regardless of the on/off functional category of these GFP mutants, as quantified by the small mean residual scores (0.09 and −0.07) for the two conservative subsets.

Expanding on the above results, computed residual scores for 13 590 multiple GFP mutants with experimentally measured fluorescence for which all residue replacements were either conservative or non-conservative were combined with those of the 1051 single residue GFP variants, to similarly elucidate a statistically significant GFP structure–function relationship (Welch’s *t*-test: $P < 0.0001$) and a plot with characteristics analogous to those for the single residue GFP variants (Fig. 2b).

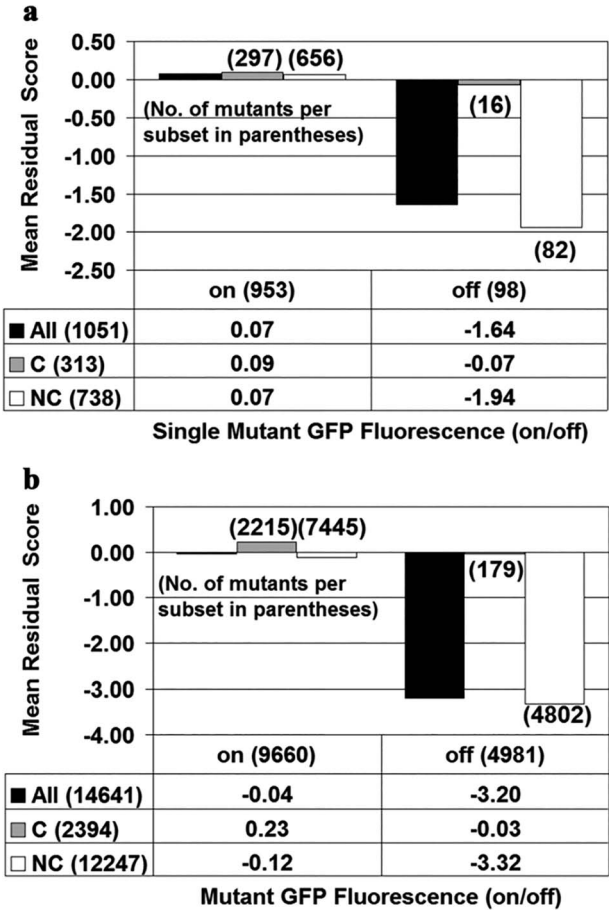


Fig. 2 GFP structure–function relationships. (a) All refers to the 1051 single residue GFP variants with experimental measured fluorescence in the data set, which are segregated into two (on/off) fluorescence categories. The C/NC subsets of variants within each category represent conservative/non-conservative amino acid substitutions of the native residue as defined in the text. The data in the table below the figure are means of the residual scores for the various sets of variants. Numbers in parentheses located either on the graph or in the table row/column headings are counts for the total number of variants in those sets. (b) Multiple mutants of GFP, for which all the residue replacements were either C) or NC, were added to the single residue GFP mutants to yield a plot similar to that observed in (a).

Residual scores cluster GFP residues by polarity and distinguish their roles

As described in the Materials and methods, a RES was calculated for each of the 229 positions in the native GFP protein, together forming a potential profile. Next, for each position, residual scores were computed for all 19 single amino acid replacements of the native residue; moreover, the mean of these 19 residual scores defines a ‘comprehensive mutational profile’ (CMP) score for each position, quantifying the average impact to sequence–structure compatibility of replacing the native amino acid with all 19 possible alternatives. Over all 229 residue positions in GFP, their respective CMP and RES values were inversely correlated ($r^2 = 0.67$) (Fig. 3a). The 19 single residue GFP variants at each position were subsequently placed into two subsets, depending on whether the mutation was a conservative (C) or non-conservative (NC) substitution of the native amino acid (see structure–function relationships section for definitions), so that separate C-CMP and NC-CMP mean residual scores could be

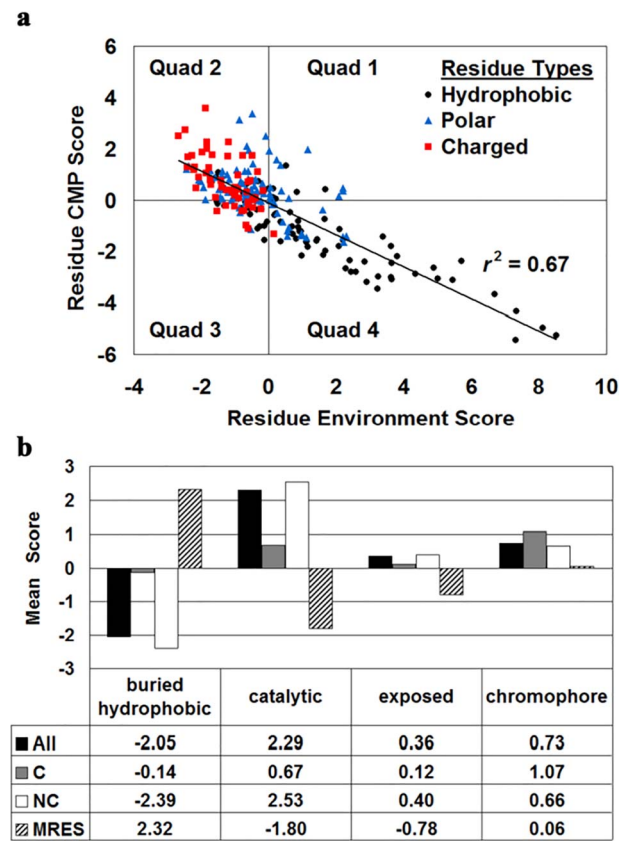


Fig. 3 (a) The CMP and RESs of the GFP amino acids are inversely correlated, with amino acids forming clusters based on their polarity. **(b)** The computational mutagenesis data distinguishes between GFP residues grouped by their structural or functional roles in the protein. All refers to the mean of the residual scores for all 19 GFP variants at all positions belonging to each group, and C/NC are the subsets of all variants in each group that represent conservative/non-conservative amino acid substitutions. MRES refers to the mean of the RESs for all the positions in each group.

calculated for each subset of variants at each position. An inverse correlation ($r^2 = 0.67$) was maintained between the NC-CMP and RES values over all GFP positions, whereas no meaningful correlation ($r^2 = 0.06$) was observed between the C-CMP and RES, reflecting the fact that non-conservative residue replacements drive the trend observed in Fig. 3a.

Furthermore, Fig. 3a shows the GFP positions forming clusters based upon polarity of the native residues, with the majority of hydrophobic and charged residues sequestered in Quadrants 4 and 2, respectively, whereas polar residues surround the origin. The plot reflects the facts that hydrophobic amino acids generally interact well with one another (i.e. positive RES), and residue replacements at those positions are detrimental to sequence-structure compatibility on average (i.e. negative CMP); that exposed charged residues generally favor the aqueous surrounding over neighboring amino acids in the structure (i.e. negative RES), and residue substitutions at those positions interact more favorably with those neighbors on average (i.e. positive CMP); and polar residues are more or less moderately ambivalent to their neighbors (i.e. RES of relatively small magnitude), with amino acid replacements at those positions having equally moderate effects on sequence-structure compatibility (i.e. CMP of relatively small magnitude). Finally, a chi-square test was applied to the distribution of all GFP residues over the (4

Quads) $\times$ (3 polarities) given by Fig. 3a, which led to the rejection of the null hypothesis that no association exists between quadrant locations and residue polarities ($\chi^2 = 84.1$, 6 degrees of freedom; $P < 0.0001$).

To investigate the capacity of the computational data to distinguish groups of GFP residue positions based on the structural or functional roles they play in the protein, we selected 46 hydrophobic residues with side chains that point toward the interior of the β -barrel (buried hydrophobic) (Chudakov *et al.*, 2010), the 2 catalytic residues (Day and Davidson, 2009), 64 exposed residues as determined by their solvent accessible surface area via the GetArea software under default conditions (Fraczkiewicz and Braun, 1998) and the 3 residues forming the chromophore. For each group of residue positions, we computed the mean of the residual scores for the 19 single residue GFP variants at all positions in the group combined (i.e. a generalized group-CMP covering multiple positions), referred to as ‘All’ in Fig. 3b. Next, all the GFP variants in each group were segregated into two subsets according to whether the mutation represented a conservative (C) or non-conservative (NC) residue replacement, and a mean of the residual scores was calculated for each subset of variants (Fig. 3b). Lastly, for each group the environment scores of all the constituent amino acids (from the potential profile of the native structure) were averaged to obtain a mean residue environment score (MRES) (Fig. 3b). The four GFP residue groups are clearly distinguished from one another by these mean scores; moreover, Fisher’s exact test was applied to the distribution of these 115 residues in a (4 groups) $\times$ (4 quads) table, which led to rejection of the null hypothesis that no association exists between the structural/functional residue groups and quadrant locations ($P < 0.0001$), given that most buried (resp. exposed) residues are found in quadrant 4 (resp. quadrant 2) of Fig. 3a.

Predictive modeling of GFP fluorescence with machine learning

All 46 310 single and multiple mutants of GFP for which fluorescence had been experimentally measured were represented by their respective 229-dimensional residual profiles, and they were randomly segregated into 10 disjoint subsets each consisting of 4631 mutants. Each subset was used to train both an RF classifier and a tree regression model, and these were subsequently used for predicting fluorescence for the GFP mutants in the other nine subsets. The evaluation of each model (trained using 10% of the data) was calculated as the prediction performance over all GFP mutants in the other subsets combined (90% of the data for testing) and reported in Table I. The predictions were relatively accurate and consistent across the distinct models trained using each of the 10 subsets. Furthermore, an examination of every training/testing pair of subsets generated equally reliable and consistent predictions, so that no particular test subset(s) biased the overall performance results of any trained model (Supplementary Data 1). This approach is related to, but clearly distinct from, a traditional tenfold cross-validation (10-fold CV) procedure, whereby (i) 10 disjoint subsets of equal size are generated from a single (training) set, (ii) one of the subsets (10% of the data) is held-out and predicted by a model trained using the set formed by combining the other nine subsets (90% of the data), and 3) the process is iterated 10 times so that each subset is held-out exactly once and predicted. A key reason for implementing this related form of 10-fold CV is that negligible improvement and a rapid plateau was observed in 10-fold CV performance for training sets marginally $>10\%$ of the GFP variant data.

Table I. RF classification and tree regression cross-validation performance^a using ten disjoint subsets (4631 mutants each). A model trained using one subset was used for predicting all mutants in the other nine subsets

Training set									
1	2	3	4	5	6	7	8	9	10
se = 0.92	se = 0.92	se = 0.93	se = 0.91	se = 0.93	se = 0.93	se = 0.92	se = 0.91	se = 0.92	se = 0.92
sp = 0.87	sp = 0.87	sp = 0.86	sp = 0.88	sp = 0.86	sp = 0.86	sp = 0.87	sp = 0.88	sp = 0.87	sp = 0.86
ppv = 0.90	ppv = 0.90	ppv = 0.89	ppv = 0.91	ppv = 0.90	ppv = 0.90	ppv = 0.90	ppv = 0.91	ppv = 0.90	ppv = 0.90
bar = 0.89	bar = 0.90	bar = 0.89	bar = 0.89	bar = 0.89	bar = 0.89	bar = 0.89	bar = 0.89	bar = 0.89	bar = 0.89
mcc = 0.79	mcc = 0.79	mcc = 0.79	mcc = 0.79	mcc = 0.79	mcc = 0.79	mcc = 0.79	mcc = 0.79	mcc = 0.79	mcc = 0.79
r = 0.82	r = 0.82	r = 0.82	r = 0.82	r = 0.83	r = 0.82	r = 0.81	r = 0.81	r = 0.81	r = 0.82

^aPerformance: RF classification (se, sp, ppv, bar and mcc) and tree regression (Pearson's correlation *r*).

Delving deeper into model performance, we randomly selected two subsets (sets 5 and 7) for further analysis. Initially, an RF classifier that was trained with set 5 was tested on set 7 (auc = 0.96, bar = 0.90, and mcc = 0.81). Next, the process was repeated using the same sets for training and testing, only after shuffling the measured on/off fluorescence labels corresponding to the GFP mutants that belong to: test set 7 only (auc = 0.50, bar = 0.50, and mcc = 0.00), training set 5 only (auc = 0.50, bar = 0.49, and mcc = -0.01) and both sets (auc = 0.50, bar = 0.50, and mcc = -0.01). The ROC curves corresponding to these four scenarios (Fig. 4a) suggest that signals embedded in the structure-based residual profiles representing the GFP mutants are clearly capable of reliably distinguishing between their on/off fluorescence (functional labels), but those signals prove to be uninformative once the labels are randomly shuffled among the mutants. For a more systematic examination of the statistical significance of prediction performance, we randomly permuted the on/off fluorescence labels corresponding to the GFP mutants of test set 7 to generate 1000 control test sets (shuffled versions). In each case, we used the RF model trained with unaltered set 5 to predict the control test set 7 mutants (bar = 0.50 ± 0.01, mcc = 0.00 ± 0.02). The distribution of bar and mcc values over the 1000 control versions of test set 7 are shown in Fig. 4b, as are arrows indicating the locations for the bar and mcc obtained using the unaltered test set 7, so that $P < 0.001$ for the predictive power of our model. Similar results were obtained based on 1000 shuffles in set 5 (bar = 0.49 ± 0.01, mcc = -0.02 ± 0.03) and in both sets (bar = 0.50 ± 0.01, mcc = 0.00 ± 0.02). Lastly, learning curves were generated for set 5 to visualize the effect of training set size on model performance (Fig. 4c). For each training set size, 10 subsets were created by randomly selecting the GFP mutants from set 5, and the RF classifier was implemented using 10-fold CV testing with each subset to evaluate mean bar, mcc, and auc performance measures and their standard deviations. The fact that the curves in Fig. 4c have not quite reached a plateau suggests that additional GFP mutants may marginally improve performance.

Single residue GFP variant analysis

In total, there exist $229 \times 19 = 4351$ possible single residue mutations of GFP, given the structure (PDB accession code 2wur, chain A) under consideration for this study. Of these mutations, 1051 were in the experimental data set, with only 98 (9.3%) of them having a deleterious effect on fluorescence (98 off/953 on). This experimental data set of single residue GFP variants, represented by their 229-dimensional residual profiles, was used to train an RF model for the purpose of predicting fluorescence categories for the remaining

3300 single residue GFP mutations, which were relegated to a test set and similarly represented by their residual profiles. Given the extreme nature of the fluorescence class imbalance among the GFP mutants in the training set, appropriate measures for training a model that displays equally accurate sensitivity and specificity include implementing cost-sensitivity during training to assume higher cost for minority class prediction errors, under-sampling the majority class in the data set before training, or over-sampling the minority class before training. We decided to apply the last approach by using synthetic minority oversampling technique (SMOTE) (Chawla *et al.*, 2002) to balance the classes in the data set before training. With this modified data set, 10-fold CV testing based on RF classification produced the following performance measures: bar = 0.95, mcc = 0.91, auc = 0.99. The RF model trained with this modified data set was then used to predict on/off fluorescence categories for the 3300 single residue GFP variants in the test set (Fig. 5).

Predictions made by the trained RF model, as presented in Fig. 5, accurately reflect known biological properties concerning GFP from experimental studies. For instance, all mutations at chromophore positions Y66 and G67 were predicted to be non-fluorescent, whereas mutations at S65 were predicted to remain fluorescent, in line with the known experimental mutations at these positions. Most mutations at the catalytic position R96 were predicted to be non-fluorescent, highlighting its more critical role in forming direct contact with the chromophore and promoting backbone cyclization (Chudakov, Matz, Lukyanov and Lukyanov, 2010). On the other hand, the E222 mutations were predicted to remain fluorescent, which mirrors mutagenesis studies demonstrating that residue replacements at this position have nearly no impact on chromophore maturation (Chudakov, Matz, Lukyanov and Lukyanov, 2010). Lastly, a prediction of non-fluorescence was made for 404 of 3300 (12.2%) of the test set mutants, which is proportionally similar to that observed in the experimental data set.

GFP multiple mutants predict single residue variants

As an alternative to relying on the highly imbalanced set of 1051 single residue GFP variants (98 off/953 on) and a SMOTE procedure to train a predictive model for the remaining 3300 single residue variants, next we considered the possibility of using the 45 259 multiple mutants of GFP with known fluorescence to similarly train models for predicting those variants. To that end, the GFP multiple mutants were randomly segregated into 10 disjoint subsets of equal size, RF classification and tree regression models were trained using each of the subsets, and these models were each used to initially

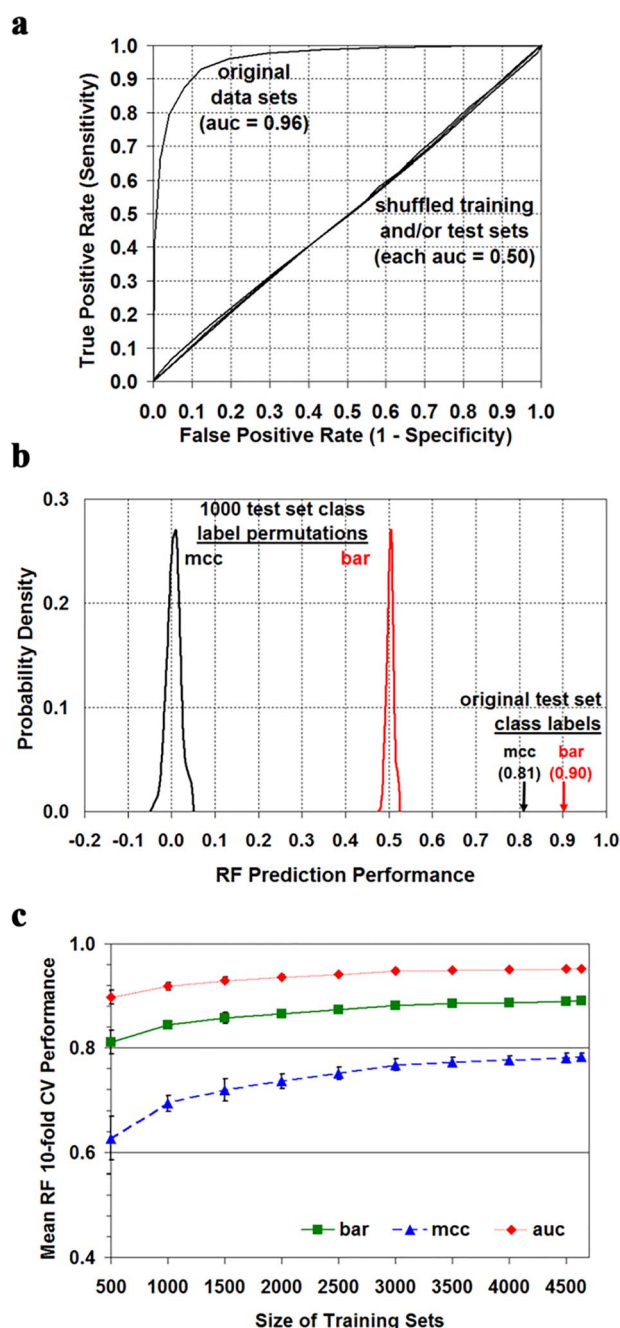


Fig. 4 (a) An ROC curve (labeled 'original data sets') obtained by predicting 4631 GFP test mutants (set 7) with an RF model trained on an independent set of 4631 GFP mutants (set 5). For comparison, three ROC curves (labeled 'shuffled training and/or test sets') were also obtained by first randomly permuting the on/off fluorescence labels of the GFP mutants in training set 5 only, test set 7 only and both sets before model training and test set prediction. (b) Distribution of bar and mcc prediction performance for the RF model, trained with the unaltered set 5, over 1000 shuffled control test sets, each obtained by randomly permuting the on/off fluorescence labels of the GFP mutants in test set 7 from their unaltered states. Also shown are the results from unaltered test set 7 (bar, balanced accuracy rate; mcc, Matthew's correlation coefficient). (c) Learning curves identify the degree to which performance may be improved with increasing numbers of GFP variants selected from set 5 for training. Each point represents the average over 10 runs of 10-fold CV, and error bars indicate one standard deviation.

predict the 1051 single residue GFP mutants to evaluate performance (Table II). The 10 RF models correctly predicted an average of 989 of 1051 (94%) of the single residue GFP mutants, with 897 of 1051 (85%) mutants being correctly predicted by all 10 models. For 1010 of 1051 (96%) of the mutants, correct predictions were made by at least 5 of 10 of the RF models; of the remaining 41 single residue GFP mutants, 14 were incorrectly predicted by all 10 RF models and 12 were incorrectly predicted by 9 of 10 RF models. Those that were difficult to predict included: mutants that introduced a backbone kink due to a proline substitution (Q184P, R109P, A179P, T43P, T225P); mutants that substituted glycine with residues having substantial side chains (G33R, G33S, G91S, G33D, G51R, G40D, G31V); and mutants that involved the replacement of buried hydrophobic or large aromatic residues (Y74C, L220Q, L44Q, V55A, Y92H, F83S, F100C, V29A, F84S). These results suggest that models trained exclusively with GFP variants harboring multiple mutations could reliably predict fluorescence classes for the remaining 3300 single residue GFP variants, with a caveat regarding the discussed limitations to the approach.

Comparisons with related methods

All 1051 single residue GFP mutants were randomly split to form a larger training set (67% of the data, 705 mutants) and a smaller test set (33% of the data, 346 mutants). Using Weka, we implemented the previously described SMOTE procedure (Chawla, Bowyer, Hall and Kegelmeyer, 2002) to balance the two class sizes for the larger set during the training of an RF classifier, which was then used to predict the test set GFP mutant class labels. Our prediction performance on the test set is shown in both charts of Fig. 6, and though we endeavored to directly compare our results with those of other state-of-the-art online tools for predicting functional consequences to protein upon single residue substitutions, there are important considerations worth noting.

Whereas our approach for representing the GFP mutants uses their residual profiles as the input attributes, related methods use various combinations of sequence, structure and evolutionary constraints as inputs to represent single residue variants. Each of the related methods is trained on a substantial collection of single residue mutations occurring in a large and diverse set of proteins, such that the same functional consequence (i.e. the output attribute) is known for all the mutants and can be predicted for new mutations in proteins with their trained model. Mutant function predictions made by these related methods are reported either categorically (e.g. neutral/no effect versus pathogenic/effect, or stabilizing versus destabilizing) or numerically (e.g. $\Delta\Delta G$ stability change values). Our training set consists solely of single residue GFP mutants, each having a categorical on/off functional consequence based on the effect of the residue replacement on the relative change to GFP fluorescence; moreover, the trained model predicts an on/off fluorescence class for held-out (i.e. unseen by the trained model) test GFP mutants. Diverse functional consequences upon mutation are generally not highly correlated on a wide scale (e.g. the direction of stability change does not clearly dictate that of fluorescence change upon all mutations at all GFP positions). Hence, direct comparison is not possible between our fluorescence class predictions on the test GFP mutants and say stability or pathogenicity predictions made by these related methods on the same test set.

Nevertheless, we sought to obtain functional predictions for the 346 GFP mutants in the test set using the following nine online methods (Fig. 6a): Pmut (Lopez-Ferrando *et al.*, 2017),

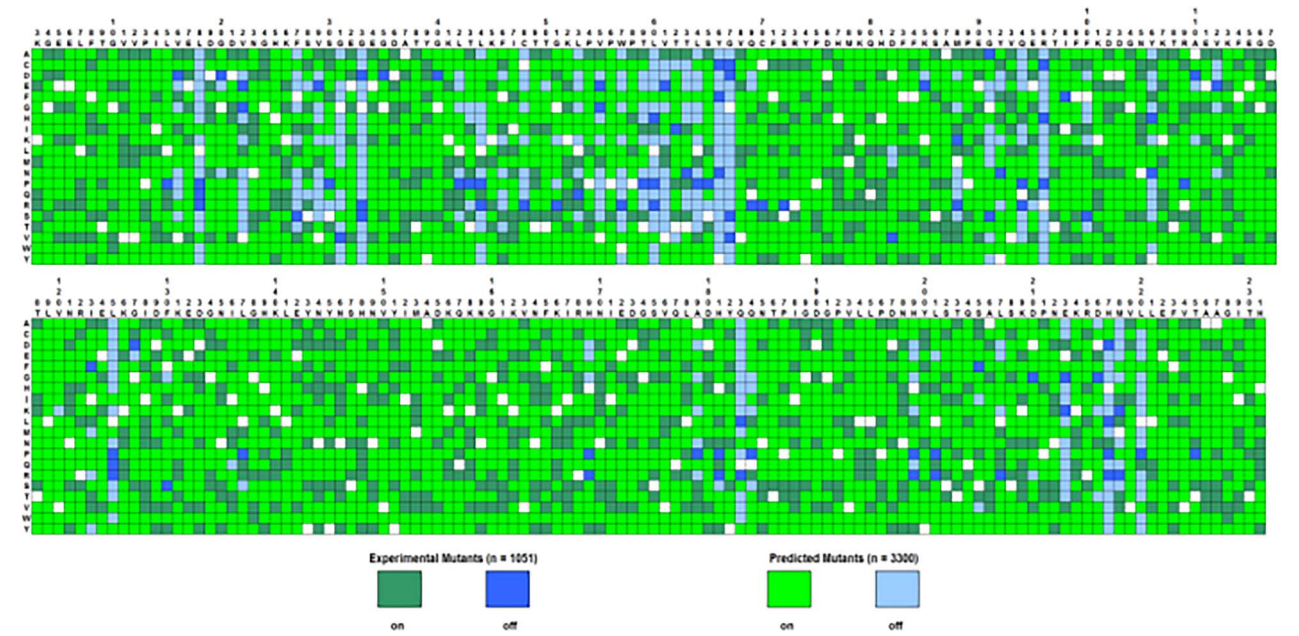


Fig. 5 Comprehensive mutational array for all single residue GFP variants. Columns represent the 229 residue positions (over two segments), with the native residue and position number above each column. Rows within each segment represent the 20 possible amino acid substitutions of the native residue, as labeled to the left of the rows. Darker colored squares represent the 1051 single residue GFP variants with experimentally known on/off fluorescence labels. Lighter colored squares represent the remaining 3300 single residue GFP mutants predicted by an RF model trained with the set of 1051 known mutants. White squares are not mutations (i.e. native and replacement residues are identical).

Table II. Multiple GFP mutants predict single mutants. Ten equal-sized disjoint subsets of the 45 259 multiple mutants were used for training RF classification and tree regression models, each of which predicted^a the test set of 1051 single GFP mutants

Training set	se	sp	ppv	bar	mcc	auc	r
1	0.96	0.76	0.97	0.86	0.68	0.93	0.67
2	0.95	0.80	0.98	0.87	0.67	0.94	0.67
3	0.96	0.71	0.97	0.84	0.64	0.95	0.68
4	0.97	0.72	0.97	0.85	0.69	0.95	0.69
5	0.96	0.70	0.97	0.83	0.64	0.93	0.69
6	0.96	0.71	0.97	0.84	0.65	0.93	0.71
7	0.97	0.73	0.97	0.85	0.69	0.94	0.69
8	0.95	0.80	0.98	0.87	0.67	0.96	0.68
9	0.97	0.65	0.96	0.81	0.65	0.94	0.68
10	0.97	0.70	0.97	0.84	0.69	0.95	0.68
avg ± stdev	0.96 ± 0.01	0.73 ± 0.04	0.97 ± 0.00	0.85 ± 0.02	0.67 ± 0.02	0.94 ± 0.01	0.68 ± 0.01

^aPerformance: RF classification (se, sp, ppv, bar and mcc) and tree regression (Pearson’s correlation *r*).

SNAP (Bromberg and Rost, 2007), SDM (Pandurangan *et al.*, 2017), CUPSAT (Parthiban *et al.*, 2007), DUET (Pires *et al.*, 2014), PoPMuSiC (Dehouck *et al.*, 2011), mCSM (Rodrigues *et al.*, 2019), Provean/SIFT (Choi and Chan, 2015) and DynaMut (Rodrigues *et al.*, 2018). The web servers for these methods required only the protein identity (either FASTA sequence or PDB accession code) and a list of single residue mutations requested for prediction as inputs. No adjustable parameters were identified with any of these tools, and if any were overlooked, then we implemented the defaults. The methods Pmut, Provean/SIFT and SNAP predicted GFP mutants in the test set to be either neutral or disrupting based on their own thresholds placed on a score that accompanied each mutant prediction. The remaining six methods all predicted $\Delta\Delta G$ stability

change values for the mutants, and for all cases except PoPMuSiC, they classified the mutants as either stabilizing ($\Delta\Delta G \geq 0$) or destabilizing ($\Delta\Delta G < 0$) (these classifications were reversed with PoPMuSiC). Applying the default thresholds indicated by these methods (i.e. neutral/stabilizing = on, disrupting/destabilizing = off) led to generally low bar and mcc performance measures over all 346 GFP mutants in the test set. Since the auc relies on actual scores or $\Delta\Delta G$ values associated with the mutant predictions and is independent of any threshold cutoff, it provided the most robust assessment. Therefore, we decided to optimize the bar and mcc values associated with each of the methods by shifting the threshold in each case, often significantly away from the default value indicated by each online server, and prediction results in

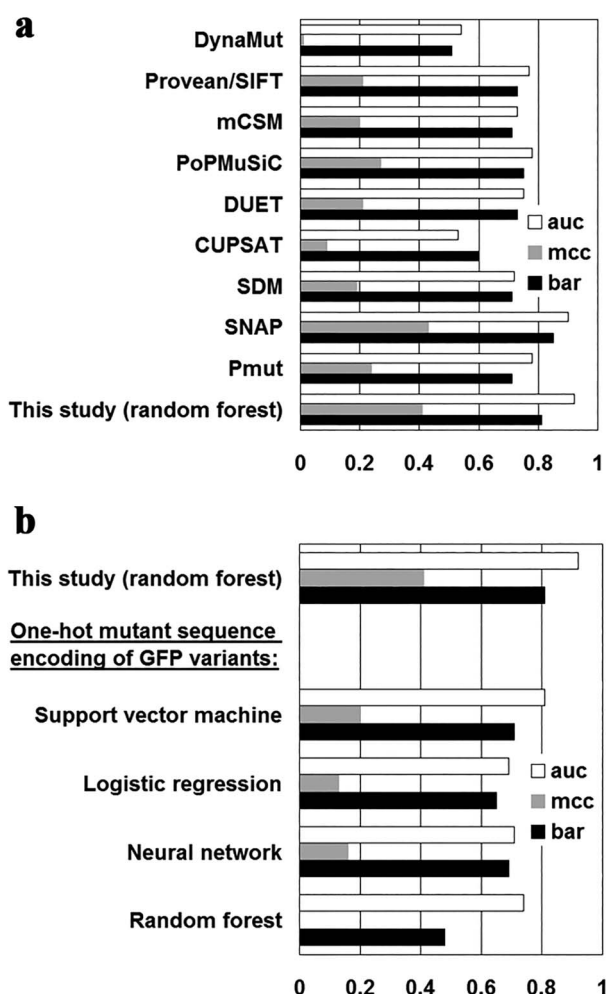


Fig. 6 Comparing model performance. (a) Single residue GFP mutant test set predictions using a trained RF model based on this study, along with predictions made on the same test set using nine state-of-the-art methods for predicting the functional effects of mutations. (b) Optimized models based on one-hot mutant sequence encodings as input attributes for the GFP variants perform no better at fluorescence class prediction than our RF model based on residual profile input attributes.

Fig. 6a for all nine methods must be considered in the context presented.

To more equitably benchmark our RF model predictions obtained by representing GFP mutants as residual profiles, we trained and evaluated a variety of optimized models based on one-hot mutant sequence encodings of the mutants as input attributes. We implemented the following algorithms in Weka with the noted changes from default parameters to optimize the models: support vector machine (SMO: buildLogisticModels = TRUE, kernel = RBF); logistic regression (SimpleLogistic: default parameters); neural network (multilayer perceptron: decay = TRUE, hiddenLayers = 2); and RF (100 iterations). We used same training/testing data sets of GFP mutants and applied the same SMOTE procedure using Weka to balance the on/off class ratio during training, identical to what was described at the start of this section for generating test set predictions by using our trained RF model. The best models (Fig. 6b) did not yield prediction performance values that fared any better than those achieved with our RF model.

Conclusions

In this study, we implemented a computational mutagenesis technique, reliant on a multibody energy function, for the purpose of characterizing the structural impacts to GFP upon single and multiple residue replacements. The scalar data calculated for thousands of GFP mutants having experimentally measured fluorescence were used to elucidate structure–function relationships in GFP, to cluster GFP residue positions by their polarity, and to distinguish between groups of GFP residues based on the structural or functional roles. Vector profile data that were also computed for the GFP mutants were used in conjunction with machine learning algorithms to train accurate and efficient predictive models of GFP fluorescence; in particular, the run-time for obtaining the computational mutagenesis data per mutant was on the order of seconds to minutes, and the time required to train the models and generate batch predictions from them was on a similar scale. Interestingly, models trained using GFP multiple mutants effectively predicted fluorescence for single residue GFP variants, highlighting the significance of the patterns of structural change signals embedded in the profiles generated through computational mutagenesis, as well as the ability of machine learning algorithms to identify and exploit those signals. These models may potentially prove useful to GFP experimentalists looking to save time and money by initially identifying a small number of promising mutagenesis leads.

Supplementary Data

Supplementary data are available at PEDS online.

Acknowledgment

We are grateful to Sarkisyan *et al.* (2016) for making available their experimental mutant GFP fluorescence data.

Conflict of Interest

The authors declare that they have no conflict of interest.

References

- Barber, C.B., Dobkin, D.P. and Huhdanpaa, H.T. (1996) *ACM Trans. Math Software*, **22**, 469–483.
- Berman, H.M., Westbrook, J., Feng, Z., Gilliland, G., Bhat, T.N., Weissig, H., Shindyalov, I.N. and Bourne, P.E. (2000) *Nucleic Acids Res.*, **28**, 235–242.
- Bowie, J.U., Luthy, R. and Eisenberg, D. (1991) *Science*, **253**, 164–170.
- Bromberg, Y. and Rost, B. (2007) *Nucleic Acids Res.*, **35**, 3823–3835.
- Chalfie, M., Tu, Y., Euskirchen, G., Ward, W.W. and Prasher, D.C. (1994) *Science*, **263**, 802–805.
- Chawla, N.V., Bowyer, K.W., Hall, L.O. and Kegelmeyer, W.P. (2002) *J Artif. Intell. Res.*, **16**, 321–357.
- Choi, Y. and Chan, A.P. (2015) *Bioinformatics*, **31**, 2745–2747.
- Chudakov, D.M., Matz, M.V., Lukyanov, S. and Lukyanov, K.A. (2010) *Physiol. Rev.*, **90**, 1103–1163.
- Cubitt, A.B., Heim, R., Adams, S.R., Boyd, A.E., Gross, L.A. and Tsien, R.Y. (1995) *Trends Biochem. Sci.*, **20**, 448–455.
- Day, R.N. and Davidson, M.W. (2009) *Chem. Soc. Rev.*, **38**, 2887–2921.
- Dayhoff, M.O., Schwartz, R.M. and Orcut, B.C. (1978) *Atlas of protein sequence and structure*, Dayhoff, M.O. (ed). National Biomedical Research Foundation, Washington, DC, pp. 345–352.
- de Berg, M., Cheong, O., van Kreveld, M. and Overmars, M. (2008) *Computational geometry: algorithms and applications*. Springer-Verlag, Berlin, Germany.

- Dehouck, Y., Kwasigroch, J.M., Gilis, D. and Rooman, M. (2011) *BMC Bioinform.*, **12**, 151.
- Fraczkiewicz, R. and Braun, W. (1998) *J. Comput. Chem.*, **19**, 319–333.
- Frank, E., Hall, M., Trigg, L., Holmes, G. and Witten, I.H. (2004) *Bioinformatics*, **20**, 2479–2481.
- Heim, R., Prasher, D.C. and Tsien, R.Y. (1994) *Proc. Natl. Acad. Sci. U. S. A.*, **91**, 12501–12504.
- Heim, R. and Tsien, R.Y. (1996) *Curr. Biol.*, **6**, 178–182.
- Leonetti, M.D., Sekine, S., Kamiyama, D., Weissman, J.S. and Huang, B. (2016) *Proc. Natl. Acad. Sci. U. S. A.*, **113**, E3501–E3508.
- Lopez-Ferrando, V., Gazzo, A., de la Cruz, X., Orozco, M. and Gelpi, J.L. (2017) *Nucleic Acids Res.*, **45**, W222–W228.
- Masso, M. and Vaisman, I.I. (2010a) *BMC Bioinform.*, **11**, 494.
- Masso, M. and Vaisman, I.I. (2007) *Bioinformatics*, **23**, 3155–3161.
- Masso, M. and Vaisman, I.I. (2008) *Bioinformatics*, **24**, 2002–2009.
- Masso, M. and Vaisman, I.I. (2010b) *PEDS*, **23**, 683–687.
- Morise, H., Shimomura, O., Johnson, F.H. and Winant, J. (1974) *Biochem.*, **13**, 2656–2662.
- Ormo, M., Cubitt, A.B., Kallio, K., Gross, L.A., Tsien, R.Y. and Remington, S.J. (1996) *Science*, **273**, 1392–1395.
- Pandurangan, A.P., Ochoa-Montano, B., Ascher, D.B. and Blundell, T.L. (2017) *Nucleic Acids Res.*, **45**, W229–W235.
- Parthiban, V., Gromiha, M.M., Abhinandan, M. and Schomburg, D. (2007) *BMC Struct. Biol.*, **7**, 54.
- Pires, D.E., Ascher, D.B. and Blundell, T.L. (2014) *Nucleic Acids Res.*, **42**, W314–W319.
- Prasher, D.C., Eckenrode, V.K., Ward, W.W., Prendergast, F.G. and Cormier, M.J. (1992) *Gene*, **111**, 229–233.
- Prasher, D.C., McCann, R.O. and Cormier, M.J. (1986) *Methods Enzymol.*, **133**, 288–298.
- Prasher, D.C., McCann, R.O., Longiaru, M. and Cormier, M.J. (1987) *Biochem.*, **26**, 1326–1332.
- Rodrigues, C.H., Pires, D.E. and Ascher, D.B. (2018) *Nucleic Acids Res.*, **46**, W350–W355.
- Rodrigues, C.H.M., Myung, Y., Pires, D.E.V. and Ascher, D.B. (2019) *Nucleic Acids Res.*, **47**, W338–W344.
- Sarkisyan, K.S., Bolotin, D.A., Meer, M.V. et al. (2016) *Nature*, **533**, 397–401.
- Shimomura, O. (1979) *FEBS Lett.*, **104**, 220–222.
- Shimomura, O., Johnson, F.H. and Saiga, Y. (1962) *J. Cell. Comp. Physiol.*, **59**, 223–239.
- Shinobu, A., Palm, G.J., Schierbeek, A.J. and Agmon, N. (2010) *J. Am. Chem. Soc.*, **132**, 11093–11102.
- Singh, R.K., Tropsha, A. and Vaisman, I.I. (1996) *J. Comput. Bio.*, **3**, 213–221.
- Sippl, M.J. (1993) *J. Comput. Aided Mol. Des.*, **7**, 473–501.
- Sippl, M.J. (1995) *Curr. Opin. Struct. Biol.*, **5**, 229–235.
- Smith, T.C. and Frank, E. (2016) *Methods Mol. Biol.*, **1418**, 353–378.
- Tsien, R.Y. (1998) *Annu. Rev. Biochem.*, **67**, 509–544.
- Wachter, R.M., King, B.A., Heim, R., Kallio, K., Tsien, R.Y., Boxer, S.G. and Remington, S.J. (1997) *Biochem.*, **36**, 9759–9765.
- Wang, G. and Dunbrack, R.L.Jr. (2003) *Bioinformatics*, **19**, 1589–1591.